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Cubic Regularized Quasi-Newton Methods

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Our Team

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Problem formulation

Convex optimization problem

$$\min_{x \in \mathbb{E}} f(x),$$

$f(x)$ is convex function with Lipschitz-continuous gradient and Hessian with constants L_1, L_2

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Lipschitz conditions

$$\|\nabla f(x) - \nabla f(y)\|_* \leq L_1 \|x - y\|$$
$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_{op} \leq L_2 \|x - y\|$$

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$$\|\nabla^2 f(x)\|_{op} \leq L_1$$

$$\|\nabla^3 f(x)\|_{op} \leq L_2$$

Model

Taylor approximation

$$\Omega_1(f, x; y) = f(x) + \langle \nabla f(x), y - x \rangle, y \in E$$

$$\Omega_2(f, x; y) = f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} \left\langle \nabla^2 f(x)(y - x), y - x \right\rangle, y \in E$$

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Upper and lower bound

$$|f(y) - \Omega_p(f, x; y)| \leq \frac{L_p}{(p+1)!} \|y - x\|^{p+1}, \quad p \in \{1, 2\}$$

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$$|f(y) - \Omega_p(f,x;y)| \leq \frac{L_p}{(p+1)!}\|y-x\|^{p+1}, \quad p \in \{1,2\}$$

Corollary

$$f(y) \leq \Omega_p(f,x;y) + \frac{L_p}{(p+1)!}\|y-x\|^{p+1}, \quad p \in \{1,2\}$$

First-order Method

Gradient Method

$$x_{k+1} = \underset{y \in \mathbb{E}}{\operatorname{argmin}} \left\{ f(x_k) + \langle \nabla f(x_k), y - x_k \rangle + \frac{L_1}{2} \|y - x_k\|^2 \right\}$$

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Step

$$\nabla f(x_k) + L_1 (x_{k+1} - x_k) = 0$$

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Explicit form

$$x_{k+1} = x_k - \tfrac{1}{L_1} \nabla f(x_k)$$

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$$x_{k+1} = x_k - \tfrac{1}{L_1} \nabla f(x_k)$$

Convergence

$$f(x_k) - f(x^*) \leq O\left(\frac{L_{\color{red}1} R^2}{k^{\color{red}1}} \right), \qquad R = \|x_0 - x^*\|$$

Second-order Method

Second-order method

$$x_{k+1} = \operatorname*{argmin}_{y \in \mathbb{E}} \left\{ f(x_k) + \langle \nabla f(x_k), y - x_k \rangle + \frac{1}{2} \left\langle \nabla^2 f(x_k)(y - x_k), y - x_k \right\rangle \right\}$$

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Newton Method [1948, L. Kantorovich]

$$x_{k+1} = x_k - \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k)$$

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$$x_{k+1} = x_k - \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k)$$

Damped Newton Method

$$x_{k+1} = x_k - h_k \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k)$$

Cubic Regularized Newton Method

Cubic Regularization [2006, Yu. Nesterov and Boris T Polyak]

$$x_{k+1} = \operatorname*{argmin}_{y \in \mathbb{E}} \left\{ f(x_k) + \langle \nabla f(x_k), y - x_k \rangle + \frac{1}{2} \left\langle \nabla^2 f(x_k)(y - x_k), y - x_k \right\rangle + \frac{L_2}{6} \|y - x_k\|^3 \right\}$$

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Step

$$\nabla f(x_k) + \nabla^2 f(x_k)(y - x_k) + \frac{L_2}{2} \|y - x_k\| (y - x_k) = 0$$

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Step

$$\nabla f(x_k) + \nabla^2 f(x_k)(y - x_k) + \frac{L_2}{2} \|y - x_k\|(y - x_k) = 0$$

Convergence

$$f(x_k) - f(x^*) \leq O\left(\frac{L_2 R^3}{k^2} \right)$$

Inexact Second-order Methods

Exact Newton Method

$$x_{k+1} = x_k - h_k \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k)$$

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BFGS [1970, C. Broyden, R. Fletcher, D. Goldfarb, D. Shanno]

$$B_{k+1} = B_k + \frac{y_k y_k^T}{y_k^T s_k} - \frac{B_k s_k s_k^T B_k^T}{s_k^T B_k s_k}$$

BFGS

$$s_k = x_{k+1} - x_k$$
$$y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$$

Inexact Cubic Regularization [2017, S. Ghadimi et.al.]

Inexactness 1

$$\nabla^2 f(x_k) \preceq B_k \preceq \nabla^2 f(x_k) + \delta I$$

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$$x_{k+1} = x_k + \operatorname*{argmin}_{h \in \mathbb{E}} \left\{ f(x_k) + \langle \nabla f(x_k), h \rangle + \frac{1}{2} \langle B_k h, h \rangle + \frac{L_2}{6} \|h\|^3 \right\}$$

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Convergence

$$f(x_k) - f(x^*) \leq O\left(\frac{\delta R^2}{k} \right) + O\left(\frac{L_2 R^3}{k^2} \right)$$

Inexact Cubic Regularization [2020, A. Agafonov et.al]

Inexactness 2

$$\nabla^2 f(x_k) - \delta I \preceq B_k \preceq \nabla^2 f(x_k) + \delta I$$

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$$f(x_k) - f(x^*) \leq O\left(\frac{\delta R^2}{k}\right) + O\left(\frac{L_2 R^3}{k^2}\right)$$

Inexact Cubic Regularization, NEW

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Convergence

$$f(x_k) - f(x^*) \leq O\left(\frac{(\delta_{up} + \delta_{low}) R^2}{k} \right) + O\left(\frac{L_2 R^3}{k^2} \right)$$

Cubic L-BFGS Method

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L-BFGS

$$B_k = B_k^m = B_k^{m-1} + \frac{y_m y_m^T}{y_m^T s_m} - \frac{B_k^m s_m (B_k^m s_m)^T}{s_m^T B_k^m s_m}$$

Subproblem solution

Subproblem's gradient

$$\nabla f(x_k) + (B_k + \delta_{up} I)h^* + \frac{L_2}{2}\|h^*\|h^* = 0$$

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Subproblem's gradient

$$\nabla f(x_k) + (B_k + \delta_{up} I)h^* + \frac{L_2}{2}\|h^*\|h^* = 0$$

Solution/Line-search intuition

The solution of the subproblem can be formulated as

$$h^* = -\left(B_k + \delta I + \tfrac{L_2}{2}\|h^*\|I\right)^{-1} \nabla f(x_k)$$

Solution

$$\|h^*\| = \arg\min_{\tau \geq 0} \left\{ \left\langle \left(B_k + \delta_{up} I + \tfrac{L}{2}\tau I\right)^{-1} \nabla f(x_k), \nabla f(x_k) \right\rangle + \tfrac{L_2}{6}\tau^2 \right\}$$

Computational complexity

Sherman-Morrison-Woodbury formula for m-rank approximation

$I^d \in \mathbb{R}^{d\times d}$, $A = \lambda I^d \in \mathbb{R}^{d\times d}$, $U \in \mathbb{R}^{d\times m}$, $V \in \mathbb{R}^{m\times d}$, $c = \nabla f(x_k) \in \mathbb{R}^d$

$$c^T(A+UV)^{-1}c = c^T A^{-1} c - c^T A^{-1} U (I^m + VU)^{-1} V A^{-1} c$$

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$$c^T(A+UV)^{-1}c = c^T A^{-1}c - c^T A^{-1}U(I^m + VU)^{-1}VA^{-1}c$$

Complexity

$$VU - \textbf{ multiplication: } O(m^2 d),$$
$$(I^m + VU)^{-1} - \textbf{ inversion: } O(m^3),$$
$$\textbf{Total: } O(m^2 d).$$

Cubic complexity

$$\textbf{SVD} \; - O(d^3)$$

Convergence Theorem

Inexactness

$$\delta_{up} = L_1, \quad \delta_{low} = mL_1, \qquad \textit{in special "greedy" version } \delta_{low} = 0$$

Theorem

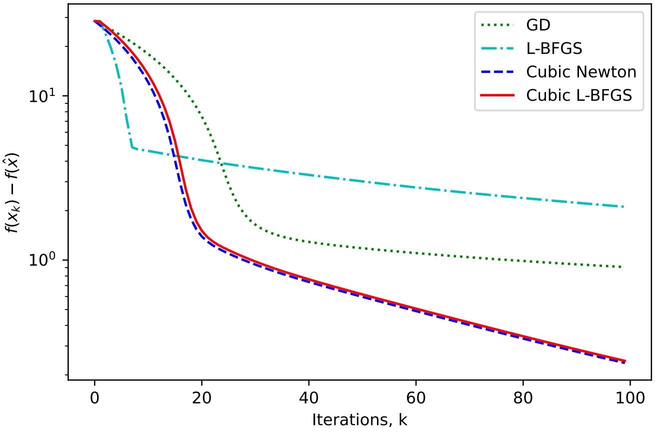
Let $f(x)$ be a convex function $f(x)$ has L_1-Lipschitz-continuous gradient and L_2-Lipschitz-continuous Hessian, B_k is an m-memory L-BFGS approximation, then the convergance rate is

$$f(x_k) - f(x^*) \leq O\left(\frac{(L_1 + mL_1)R^2}{k} + \frac{L_2 R^3}{k^2}\right),$$

where $R = \|x_0 - x^*\|$.

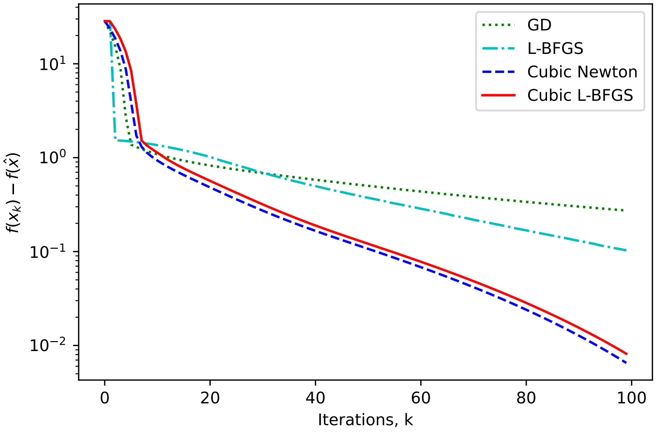
Experiments

Figure: Logistic regression, dataset = a9a, $\mu = 10^{-4}$, $m = 10$, $L_1 = 0.25$, $lr = \mu/(L_1^2)$, $L_2 = 0.1, x_0 = 10 \cdot e$



Experiments

Figure: Logistic regression, dataset = a9a, $\mu = 10^{-4}$, $m = 10$, $L_1 = 0.04$, $lr = 0.0123$ and $L_2 = 0.011, x_0 = 10 \cdot e$.



Contacts

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